

LAB1
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Community Tool & Equipment Library — Requirements Table

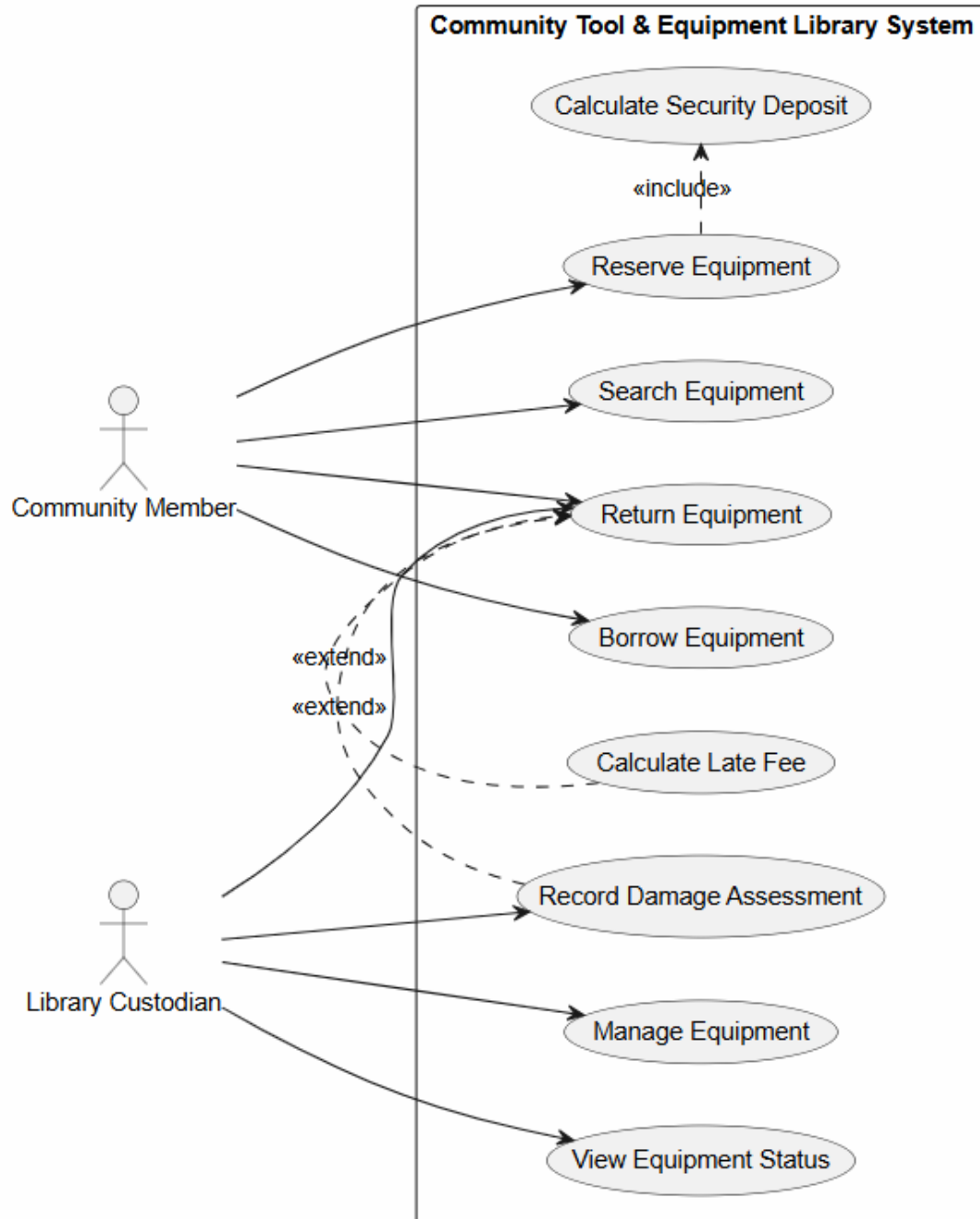
ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow community members to search for equipment by category and availability.	High	Pass: Matching equipment and availability are displayed. Fail: Incorrect availability is displayed.	Helps members quickly find required equipment.
FR-002	Functional	The system shall allow community members to reserve available equipment for a specified loan period.	High	Pass: Reservation is confirmed when equipment is available. Fail: Unavailable equipment can be reserved.	Prevents conflicts between members.
FR-003	Functional	The system shall calculate the required security deposit based on the equipment category.	High	Pass: Correct deposit is calculated before reservation. Fail: Incorrect or missing deposit.	Protects the library against equipment loss or damage.
FR-004	Functional	The system shall calculate late return fees based on the duration of the overdue period.	High	Pass: Correct late fee is calculated for an overdue return. Fail: Fee is missing or incorrect.	Encourages members to return equipment on time.
FR-005	Functional	The system shall allow the library custodian to record damage assessments when equipment is returned.	Medium	Pass: Damage details are recorded and equipment condition is updated. Fail: Damage information cannot be recorded.	Maintains accurate equipment condition records.
NFR-001	Performance & Security	The system shall display real-time physical availability of equipment across neighborhood depot lockers.	High	Pass: Current availability is accurate during simulated peak usage. Fail: Availability is outdated or incorrect.	Provides reliable availability information before reservations.
NFR-002	Security	The system shall restrict equipment management and damage-assessment functions according to the user's role.	High	Pass: Community members cannot access custodian-only functions. Fail: Unauthorized users can modify protected records.	Prevents unauthorized modification of equipment and damage records.

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Community Tool & Equipment Library - Use Case Diagram



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Community Tool & Equipment Library

Use-Case Flow Specification — Reserve Equipment

Primary Actor

Community Member

Preconditions

1. The Community Member has a registered account.
2. The Community Member is logged in.
3. The selected equipment is available.
4. The member does not have an overdue equipment loan.

Postconditions

1. The equipment is reserved for the member.
2. The required security deposit is calculated.
3. The reservation details are recorded.
4. The equipment availability status is updated.

Main Success Scenario

1. Member searches for required equipment.
2. System displays matching equipment.
3. Member selects an available item.
4. System displays the equipment details.
5. Member requests a reservation.
6. System calculates the required security deposit.
7. System verifies that the member is eligible to borrow.
8. Member confirms the reservation.
9. System records the reservation.
10. System updates the equipment availability.
11. System displays the reservation confirmation.

Alternate Flow — Equipment Unavailable

If the selected equipment is no longer available, the system informs the member that the equipment cannot be reserved and returns the member to the equipment search.